Unit 2.1 – Graphical Techniques, Box Plot, Skewness and Kurtosis, Descriptive Statistics

1. Graphical Techniques

Definition

Graphical Techniques are visual methods used to represent data in the form of graphs or charts. They help in understanding patterns, trends, comparisons, and distributions quickly.

Importance

- Makes data easy to understand.
- Identifies trends and patterns.
- Compares different datasets.
- Helps in decision-making.
- Detects outliers and unusual values.

Types of Graphical Techniques

A) Bar Chart

- Represents categorical data using rectangular bars.
- Height of each bar shows the value.

Example: Number of students in different classes.

B) Line Chart

- Displays data points connected by lines.
- Shows changes over time.

Example: Monthly sales of a company.

C) Pie Chart

- Represents data as slices of a circle.
- Shows the percentage contribution of each category.

Example: Family monthly expenditure.

D) Histogram

- Represents the frequency distribution of continuous data.
- Bars touch each other because the data is continuous.

Example: Distribution of students' marks.

E) Scatter Plot

- Displays the relationship between two numerical variables.
- Each point represents one observation.

Example: Height vs Weight of students.

Summary Table

Graph	Purpose	Example
Bar Chart	Compare categories	Students in each class
Line Chart	Show trends over time	Monthly sales
Pie Chart	Show percentages	Budget distribution
Histogram	Frequency distribution	Marks of students
Scatter Plot	Relationship between variables	Height vs Weight

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2. Box Plot

Definition

A **Box Plot (Box-and-Whisker Plot)** is a graphical representation used to show the distribution of data based on the **five-number summary**.

Five-Number Summary

1. Minimum Value
2. First Quartile (Q1)
3. Median (Q2)
4. Third Quartile (Q3)
5. Maximum Value

Components of a Box Plot

- **Box:** Represents the middle 50% of the data (Q1 to Q3).
- **Median Line:** A line inside the box showing the median.
- **Whiskers:** Extend from the box to the minimum and maximum values.
- **Outliers:** Data values that lie far from the rest of the data.

Advantages

- Easy to compare datasets.
- Identifies outliers.
- Shows spread and variability.
- Displays symmetry or skewness.

Example

Marks: 20, 30, 40, 50, 60, 70, 80

Minimum = 20

Q1 = 30

Median = 50

Q3 = 70

Maximum = 80

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3. Skewness

Definition

Skewness is a measure of the **asymmetry** of a data distribution.

It tells whether the data is balanced or tilted to one side.

Types of Skewness

A) Positive (Right) Skewness

- Tail is longer on the **right side**.
- Mean > Median > Mode.

Example: Most students score low marks, but a few score very high.

B) Negative (Left) Skewness

- Tail is longer on the **left side**.
- Mean < Median < Mode.

Example: Most students score high marks, but a few score very low.

C) Zero Skewness (Symmetrical Distribution)

- Both sides are equal.
- Mean = Median = Mode.

Example: A perfectly balanced distribution.

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4. Kurtosis

Definition

Kurtosis measures the **peakedness or flatness** of a data distribution.

It shows how concentrated the data is around the mean.

Types of Kurtosis

A) Mesokurtic

- Normal distribution.
- Moderate peak.

B) Leptokurtic

- High and sharp peak.
- More data concentrated near the mean.

C) Platykurtic

- Flat and broad peak.
- Data is more spread out.

Summary Table

Type	Shape	Characteristics
Mesokurtic	Normal	Moderate peak
Leptokurtic	Sharp	High peak, less spread
Platykurtic	Flat	Low peak, more spread

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5. Descriptive Statistics

Definition

Descriptive Statistics is the branch of statistics used to **collect, organize, summarize, and present data** in a meaningful way.

It describes the main features of a dataset without making predictions.

Objectives

- Summarize large amounts of data.
- Present data in a simple form.
- Find central values.
- Measure variability.
- Support decision-making.

Measures Used in Descriptive Statistics

A) Measures of Central Tendency

- Mean
- Median
- Mode

B) Measures of Dispersion

- Range
- Variance
- Standard Deviation

C) Graphical Representation

- Bar Chart
- Pie Chart
- Histogram

- Box Plot
- Line Chart
- Scatter Plot

Difference Between Descriptive and Inferential Statistics

Descriptive Statistics	Inferential Statistics
Summarizes existing data	Predicts or draws conclusions about a population
Uses graphs and averages	Uses probability and hypothesis testing
No prediction	Makes predictions and decisions
Example: Average marks of a class	Example: Predicting election results from a sample

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Important 2-Mark Questions

1. Define Graphical Techniques.
2. What is a Box Plot?
3. Define Skewness.
4. What is Positive Skewness?
5. What is Negative Skewness?
6. Define Kurtosis.
7. What is Descriptive Statistics?
8. State any two graphical techniques.
9. What are outliers?
10. List the five-number summary used in a box plot.

Important 4/6-Mark Questions

1. Explain Graphical Techniques with suitable examples.
2. Explain the Box Plot with a neat diagram.
3. Explain the types of Skewness with examples.
4. Explain the types of Kurtosis.
5. Explain Descriptive Statistics and its measures.
6. Differentiate between Descriptive Statistics and Inferential Statistics.